BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQ's)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. Number of electrons in a shell is a given by:

(a) 2n^3
(b) 2n^2
(c) 2(2l+1)
(d) 2l2

Q2. Oxidation number of oxygen in OF2 is:

(a) -2
(b) -1
(c) +1
(d) +2

Q3. The shape of p-orbital is:

(a) Spherical
(b) Complex
(c) Dumb-bell
(d) double dumb-bell

Q4. The crystal of sodium chloride is made up of _____ unit cells.

(a) Trigonal
(b) Cubic
(c) orthrhombic
(d) Monoclinic

Q5. Which orbital has lowest energy?

(a) 3d
(b) 4s
(c) 3p
(d) 4f

Q6. If dx/dl = K then concentration of each reactant in a reaction must be:

(a) 2M
(b) 3M
(c) 10M
(d) 1M

Q7. Mass of electron is:

(a) 9.11 x 10^-28g
(b) 9.11 x 10^-31Kg
(c) 0.000055 a.m.u
(d) All of these

Q8. Neutron was discovered by:

(a) W. Crookes
(b) Gold Stein
(c) Moseley
(d) James Chadwick

Q9. Which of the following figure contain three significant digits:

(a) 1.030
(b) 0.035
(c) 0.0010
(d) 1030

Q10. Which of the following molecule posses highest diplomoment?

(a) CO
(b) CO2
(c) CS2
(d) BeCl2

Q11. How many moles are there in 8 g of hydrogen gas:

(a) 2 moles
(b) 4 moles
(c) 6 moles
(d) 8 moles

Q12. 16 g of oxygen gas at S.T.P occupies:

(a) 11.2 dm^3
(b) 22.4 dm^3
(c) 44.8 dm^3
(d) 67. dm^3

Q13. The dissociation of an electrolyte depends upon:

(a) Nature of electrolyte
(b) Degree of dilution
(c) Temperature
(d) All of these

Q14. For the decomposition of H2O2, the inhibitor is:

(a) MnO2
(b) V2O5
(c) Glycerine
(d) Al2O3

Q15. Alpha particle consist of :

(a) 2 protons and 2 neutrons
(b) 3 protons and 2 electrons
(c) 1 proton and 2 neutron
(d) 2 protons and 1 neutron

Q16. Second energy level contain:

(a) Two sub- shells
(b) Four orbitals
(c) Maximum eight electrons
(d) All of these

Q17. Oxidation number of oxygen in KO2 is:

(a) -2
(b) -1
(c) -1/2
(d) +2

Q18. Strongest sigma bond is formed due to _____ orbital overlapping.

(a) s - s
(b) s - p
(c) p - p
(d) All of these

Q19. When 1 mole of a compound is formed at 25oC and 1 atm pressure then it is called:

(a) HF
(b) Hf
(c) H
(d) E

Q20. Which crystalline solid posses low melting point and low density?

(a) Atomic solid
(b) Molecular solid
(c) Ionic solid
(d) Metallic solid

Q21. The total number of electrons in a sub-shell having l=2 are:

(a) 2
(b) 6
(c) 8
(d) 10

Q22. Which one of the following is an element?

(a) Sea water
(b) Graphite
(c) Air
(d) Soil

Q23. The volume of gas varies with:

(a) Temperature
(b) Pressure
(c) Moles
(d) All of these

Q24. Phenolphthalein indicator given _____ color in basic solution.

(a) Yellow
(b) Green
(c) Pink
(d) Blue

Q25. Which one of the following has zero depolemoment?

(a) CH4
(b) NH3
(c) HF
(d) H2O

Q26. The law of chemical equilibrium is given by:

(a) Le- Chatelier
(b) Guldberg and Waage
(c) Bohr
(d) Einstein

Q27. According to _____ Law Pv = K at constant temperature.

(a) Boyle's
(b) Charle's
(c) Avogadro's
(d) Graham's

Q28. The rate of diffusion / effusion of gases depends upon their:

(a) Molecular Mass
(b) Molecular Motion
(c) Both (a) and (b)
(d) None of these

Q29. Which one of the following can not be hydrolyzed?

(a) NaCl
(b) NH4Cl
(c) Na2CO3
(d) Both (b) and (c)

Q30. Bond energy is maximum in case of:

(a) Polar Bond
(b) None-polor Bond
(c) Both (a) and (b)
(d) Noen of these

Q31. Rate of formation of product for a chemical reation A → B can be denoted as :

(a) dt/dx
(b) -dA/dt
(c) dB/dt
(d) -dt/dx

Q32. e/m ratio of electron is determine by:

(a) William Crookes
(b) Gold Stien
(c) Milliken
(d) J.J Thomson

Q33. The grading of motor oil is done on the basis of its:

(a) Boiling point
(b) Vapour pressure
(c) Viscosity
(d) Surface tension

Q34. One atomic unit is equal to 1/12 th of the mass of:

(a) C12
(b) C13
(c) C14
(d) All of these

Q35. The S.I Unit of dipole-moment is:

(a) Debye
(b) esu cm
(c) Coloumb meter
(d) Joule

Q36. The unit of rate constant for zero order reaction is:

(a) s-1
(b) mol l-1s-1
(c) l2 mol-2 s-1
(d) 1 mol-1 s-1

Q37. If value of l = 3 then orbital is:

(a) s
(b) p
(c) d
(d) f

Q38. Isomorphism is the property of:

(a) Crystalline solids
(b) Amorphons solids
(c) Liquids
(d) Gases

Q39. Rage of reaction is directly proportional to:

(a) Concentration of Reactant
(b) Concentration of Product
(c) Inhibitor
(d) Order of Reaction

Q40. Hydrolysis may be the reverse of:

(a) Hydration
(b) Neutralization
(c) Ionization
(d) None of these

Q41. The ability of an ion to be hydrated, depends upon its:

(a) Charge
(b) Size
(c) Both (a) and (b)
(d) None of these

Q42. At equilibrium state the composition of equilibrium mixture:

(a) [illegible in source scan]
(b) [illegible in source scan]
(c) Continuously changes
(d) Remains constant

Q43. Vapour pressure of liquid is measured by:

(a) Viscometer
(b) Manometer
(c) Stalagmometer
(d) Calorimeter

Section-B Short Answer

> Note: Answer any SIX of the following questions.

Q2. Define the following with examples: (i) Empirical formula (ii) Polymorphism

Q3. At 20oC, 480 cm^3 of a gas is kept is a vessel at a pressure fo 2.5 atm. What will be the volume of gas, if the pressure is changed to 900 toor?

Q4. Differentiate between Sigma and Pi Bond.

Q5. Describe Postulates of Bohr's theory.

Q6. Define with examples: (a) Exothermic reaction (b) Endothrmic reaction

Q7. What is hybridization? Explain sp^3 hybridization.

Q8. Define the following terms: (i) Rate of reaction (ii) Velocity of reaction

Q9. Write a note on any one of the following: (i) Rate of reaction (ii) Velocity pressure

Q10. Differentiate between oxidation and reduction reactions.

Section-C (Descriptive Answer)

> Note: Answer any TWO of the following questions.

Q11. State Le Chatelier's principle, Describe the effect of increase in concentration, pressure and temperature on the following reaction. N2(g) + 3H2(g) <=> 2NH3(g) ΔH = Negative

Q12. Define oxidation number, Balance the following redox equations by Ion-electron method: (i) Fe2+ + Cr2O7^2- + H+ → Fe3+ + Cr3+ + H2O (ii) Cr(OH)3 + H2O2 → CrO4^2- + H2O (Basic)

Q13. (a) Define following terms: (i) Radioactivity (ii) Molecular formula (b) Ethylene Glycol contains Carbon, hydrogen and Oxygen. Its molecular mass is 62 a.m.u. Combustion of 6.38 mg of ethylene Glycol gives 9.06 mg of CO2 and 5.58 mg of H2O. Find its Empirical formula and molecular formula.